VAISHNAVI BHARADWAJ

Advanced Application Engineering Analyst

Data Analyst — Product Analytics & Business Intelligence

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Portfolio



EDUCATION

Year	Degree	CGPA	Institute
2022	B.Tech (Electronics and Instrumentation)	9.14	Haldia Institute Of Technology, West Bengal
2026	MBA (Business Analytics)	9.33	Manipal Academy of Higher Education, Karnataka

WORK EXPERIENCE

Accenture – Advanced Application Engineering Analyst

Project: AT&T – Telecommunications, US (Infra, API Monitoring, Production Support) **[Jul 2023 – Dec 2025]**

- Enhanced Linux/Unix server reliability via health checks, log diagnostics, and automated service recovery, improving uptime by approx. **15%+**.
- Engineered **C2W** workflows & API observability using AppDynamics, Dynatrace, Grafana, improving alert triaging and reducing MTTR by **25%**.
- Resolved **200+** kiosk tickets, achieving **30%** faster turnaround; delivered AT&T flagship (iPhone & Samsung) sales by **Power BI** KPI dashboards.

Project: ALLIANZ - Insurance, Spain (Backend Development) **[Jan 2026 – Present]**

- Spearheaded full-cycle backend development for an **insurance application** using **PHP Symfony** in VS Code, engineering structured controller, service, and DTO layers integrated with **D3 Pick** MultiValue databases.
- Architected **Multiasistencia** integrations to optimize screening pipelines, leveraging indexed structures & query tuning to improve throughput.

PUBLICATIONS

- **Bharadwaj, V.** "Conversational Custodians: Deploying LLM-Powered Agentic Systems for Dynamic Digitization and Contextual Revival of Intangible Cultural Heritage". *Proceedings of the International Conference on Computation and Humanities: Overlapping Research in Digital Spaces (CHORDS) 2026*. Springer. **(Accepted for publication)**. Peer-reviewed. | LLM agent framework. | Certificate

PROJECTS

Data Science Learning Hub | **Full Stack** | **Demo** **[Aug 2025 - Present]**

- Built **Ed-Tech SPA** across **12** subjects and **70+** modules; **15+** visualisers covering Minimax, Alpha-Beta, K-Means, SVM, PCA, BFS-DFS, etc.
- Built **adaptive quiz engine** with spaced-repetition mastery scoring, exponential time-decay weighting, and offline IndexedDB progress tracking.
- Delivered full GATE DA syllabus with KaTeX LaTeX rendering, live Mermaid diagram generation & authentic timed auto-submit exam simulation.
- Architected **offline-first code-split** SPA with all tool bundles lazy-loaded on demand, delivering instant startup without any server dependency.

Computer Vision Research — Height Estimation **[Mar 2022 – May 2022]**

- Designed an end-to-end deep learning framework to accurately estimate the tree trunk DBH from the high-resolution field video streams.
- Screened out 95+% of motion-blurred frames via Laplacian variance filtering prior to Segment Anything Model-based instance segmentation.
- Enhanced SAM boundary precision by integrating custom PCA outline extraction with the confidence scoring and area filtering.
- Evaluated EVP depth estimation models and the integrated IMU-assisted stereo vision feeds to minimize distance errors.

Socio-Economic Health Analytics — Satellite Remote Sensing & ML **[May 2023 – Sep 2023]**

- Built predictive gradient boosting pipelines on multi-spectral satellite data, performing EDA for an 11.39 RMSE baseline.
- Isolated top 200/500 features via XGBoost importance scoring to improve spatial generalization, reducing RMSE to 11.25.
- Addressed sparse spatial data using iterative mean/median imputation, fine-tuning LightGBM to a final RMSE of 11.60.

INTERNSHIP

Java Full Stack + GCP Training | **Wipro** **[Apr 2022 – Sep 2022]**

- Completed full-stack training covering backend services, frontend frameworks, cloud workflows, and modern software development practices.
- Integrated Teachmint Admin Portal with EventBeep using GCP Functions, Cloud Storage, enabling smooth data handling across both platforms.

Automation Projects | **Cognizant** **[May 2022 - Aug 2022]**

- Built UI automation suites using Selenium + Java for the Live Support GSD Portal, Display Bookshelves app, and Online Book Search platform.
- Improved test coverage by **11%** & reduced manual regression effort through automated workflows, enabling faster and reliable release cycles.

ACHIEVEMENTS

- Served as **Joint Secretary** for the ISA and ISOI student chapters in 2021, contributing actively to events and organizational initiatives.
- Secured **1st** place in the **CIRCUITRIX** hardware competition organized by the **IEEE** Society in 2019, demonstrating strong technical skills.
- Completed a **Senior Diploma** in **Kathak** from Allahabad University in 2017, showcasing dedication to classical dance training.

COURSEWORK INFORMATION

- **Business Analytics (MBA):** Data Visualization, Exploratory Data Analysis, Regression, Business Analytics, Data governance and stackholders
- **Electronics & Instrumentation Engineering:** Microprocessor 8085 and 8086 | VLSI | Digital Computing | Analog | Optical instrumentation
- **Mathematics:** Probability & Statistics, Linear Algebra, Numerical & Complex Analysis, Transform Calculus, Advanced Calculus
- **MOOC:** Google Cloud Big Data & Machine Learning Fundamentals | Blockchain, Solidity & Smart Contracts | Data Security & Code of Ethics

SKILLS AND EXPERTISE

- **Programming Languages:** Java, JavaScript, TypeScript, Python, SQL, C, C++, R, C#, Kotlin, Swift, MATLAB, Bash, HTML
- **Frameworks & Libraries:** React, Vite, Tailwind CSS, KaTeX, Angular, Spring Boot, Express, Node.js, Selenium WebDriver
- **Cloud & DevOps:** Google Cloud Platform, Azure, Linux/Unix Administration, AppDynamics, Dynatrace, Grafana, Microservices, REST APIs
- **Developer Tools:** Power BI, VS Code, Git, GitHub, Postman, Tableau, JIRA, Automation Scripting (Shell/Python), MySQL Workbench
- **Core Competencies:** Production Support, API Monitoring, Server Operations, Troubleshooting & RCA, UI Automation, Documentation

CERTIFICATIONS

Salesforce Certified Administrator | Microsoft Azure Fundamentals | GCP ACE certification | GCP Digital Cloud Leader | MongoDB SI Associate Certification | Introduction to Internet Of Things by NPTEL | Oracle Fusion AI Agent Studio Foundations Associate | Oracle AI Foundations Associate | Machine Learning and Deep Learning Specialization by Andrew Ng, Coursera | Introduction to Algorithms, MIT OpenCourseWare